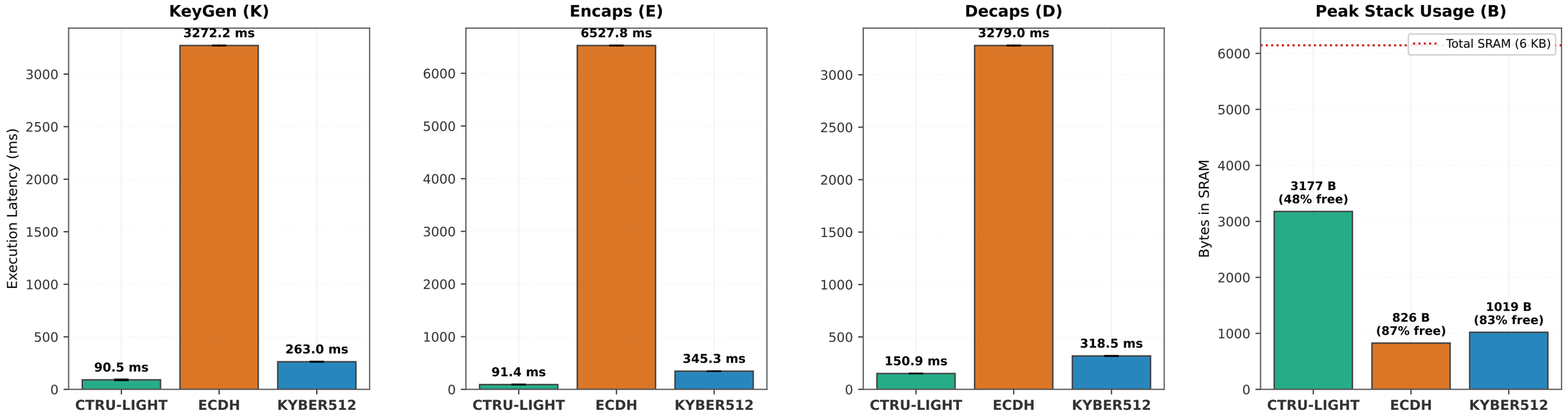
Cross-Algorithm Performance & Memory Benchmark on Physical ATmega4809 (20 MHz, ASM, -O3)
Direct hardware comparison between Post-Quantum KEMs (Kyber-512, CTRU-Light) and Classical ECDH (secp256r1 / micro-ecc).
KeyGen (K)
Execution Latency (ms)
3272.2 ms
90.5 ms
263.0 ms
Encaps (E)
6527.8 ms
91.4 ms
345.3 ms
Decaps (D)
3279.0 ms
150.9 ms
318.5 ms
CTRU-LIGHT
ECDH
KYBER512
Peak Stack Usage (B)
Bytes in SRAM
Total SRAM (6 KB)
3177 B
(48% free)
826 B
(87% free)
1019 B
(83% free)